# Statistics

— covariance & correlation

## Intuition

Variance & S.D tells us how spread out of a single column

How much does this one variable vary around its mean ?

But they say absolutely nothing about 2 variables are related

1/100 → 1% 10

Course

| X | Y | X | Y |
|---|---|---|---|
| body weight | running speed | class | completeness |
| 50kg | 40km/hr | 1 | 0 |
| 70kg | 25 km/hr | 5 | 5% |
| 100kg | 15 km/hr | 10 | 10% |
| | | 95 | 99% |

X ↑ Y ↑

X ↓ Y ↓

X ↑ Y ↓

**Covariance:** It is a statistical tool that two variables (columns) & return a single number

* is the number positive ⟶ directly proportional (both move in same direction
* is the number around zero ⟶ No relation
* is the number negative ⟶ inversely proportional (they move in opposite direction]

It is a measure of how much two random variables change together

∴ [Quantify the relationship b/w X & Y]

eg:

Dataset X & Y

①

| Rented | No. rooms | Size of house X | Price Y |
|---|---|---|---|
| | | 1200 sq.ft | 45 lakhs |
| | | 1300 sq.ft | 50 lakhs |
| | | 1500 sq.ft | 75 lakhs |

X ↑ Y ↑

X ↓ Y ↓

∴ Here if size increase ⟶ price increases

Possible relationships

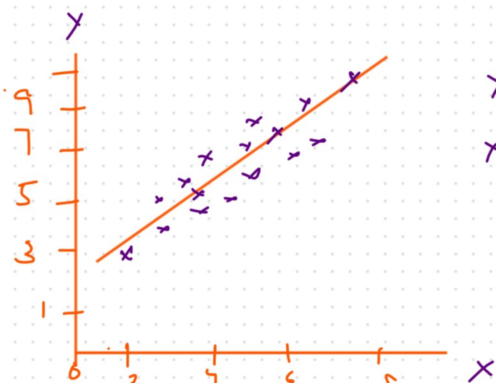
$$\boxed{\begin{array}{ll} X\uparrow & Y\uparrow \\ X\downarrow & Y\downarrow \\ X\uparrow & Y\downarrow \\ X\downarrow & Y\uparrow \end{array}}$$

} +ve covariance
+ve covariance

} −ve covariance
−ve covariance

eg ①

| X | Y |
|---|---|
| 2 | 3 |
| 4 | 5 |
| 6 | 7 |
| 8 | 9 |




Scatter plot

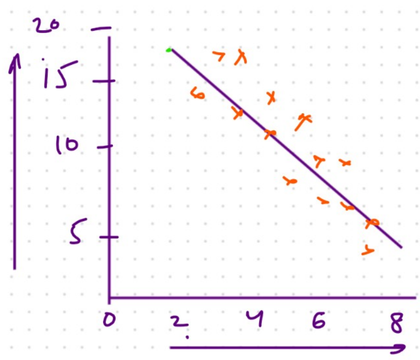
X ↑ Y ↑

X ↓ Y ↓

Positive Covariance : points form line going from bottom-left to top right

②

| x | y |
|---|---|
| 7 | 10 |
| 6 | 12 |
| 5 | 14 |
| 4 | 16 |



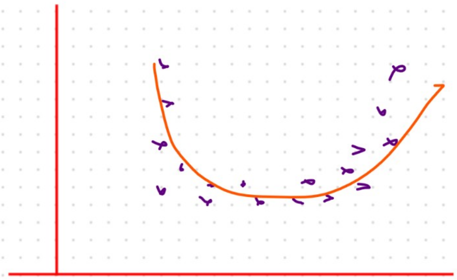

X ↑ Y ↓

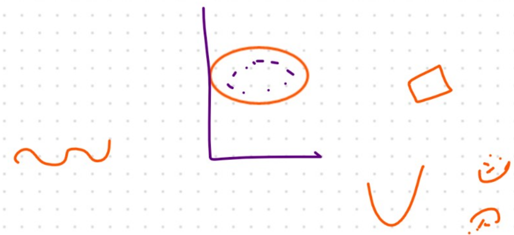
X ↓ Y ↑

−ve Coriance

line goes from top-left to bottom-right

③





Formulae for Covariance :

$$\boxed{Cov(x,y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}}$$

when quantifying relation b/w X & Y

$$Cov(x,x) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(x_i - \bar{x})}{n-1}$$

$$= \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

$$Cov(x,x) = Var(x)$$

∴ sample variance of x is nothing but Covariance of x & x

Pb Students

| Hours Studied (X) | Exam Score (Y) |
| --- | --- |
| 2 | 50 |
| 3 | 60 |
| 4 | 70 |
| 5 | 80 |
| 6 | 90 |

X ↑ Y ↑

X ↓ Y ↓

∴ +ve Covariance

Apply formulas

① $\bar{x} = \dfrac{2+3+4+5+6}{5} = 4$

② $\bar{y} = \dfrac{50+60+70+80+90}{5} = 70$

$$Cov(x,y) = \sum_{i=1}^{n} \frac{(x_i-\bar{x})(y_i-\bar{y})}{n-1}$$

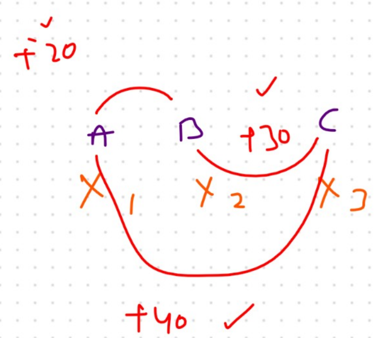
$$= \frac{(2-4)(50-70) + (3-4)(60-70) + (4-4)(70-70) + (5-4)(80-70) + (6-4)(90-70)}{4}$$

$Cov(x,y) = 20$

∴ The positive covariance indicates the no. of hours studied increased the exam score also

Advantages

① Quantify the relationship b/w x & y




Disadvantages

① Covariance doesn't have a specific unit value

∴ $Cov(x,y) \to -\infty$ to $\infty$

| Cov (A,B) | Cov (B,C) |
|---|---|
| 20 | 30 |
| 100 | 300 |
| −200 | −100 |

No limit or control to restrict

Correlation → $\boxed{-1 \text{ to } +1}$

will solve this

| Cor (A,B) | Cov (B,C) |
|---|---|
| 0.96 | 0.98 |

## Correlation:

① pearson Correlation Coefficient

② Spearman Rank Correlation

① Pearson Correlation Coefficient: [-1 to +1]

$$\boxed{\rho_{x,y} = \frac{Cov(x,y)}{\sigma_x \cdot \sigma_y}}$$

row

$\sigma_x \quad \sum_{i=1}^{n} \frac{(x_i - 4)^2}{N}$

| | | | |
|---|---|---|---|
| 2 | 4 | $= 2^2$ | $= 4$ |
| 3 | 4 | $= 1^2$ | $= 1$ |
| 4 | 4 | $= 0^2$ | $= 0$ |
| 5 | 4 | $= 1^2$ | $= 1$ |
| 6 | 4 | $= 2^2$ | $= 4$ |
| | | | 10 |

$\frac{10}{5} = 2$

$\sigma_y =$

| | | | |
|---|---|---|---|
| 50 | 70 | $= 20^2$ | $= 400$ |
| 60 | 70 | $= 10^2$ | $= 100$ |
| 70 | 70 | $= 0^2$ | $= 0$ |
| 80 | 70 | $= 10^2$ | $= 100$ |
| 90 | 70 | $= 20^2$ | $= 400$ |
| | | | 1000 |

$\frac{1000}{5} = 200$

$$\rho_{x,y} = \frac{Cov(x,y)}{\sigma_x \cdot \sigma_y} = \frac{20}{\sqrt{2} \cdot \sqrt{200}} = \frac{20}{\sqrt{400}} = \frac{20}{20} = 1$$

Note:

→ The more the value towards +1 the more the correlated x & y is

→ the more the value towards -1 the more -ve correlated is
is (x,y)

② Spearman Rank Correlation: [1]

above pearson correlation contain correlation for non linear data like 0.88 not 1, but this can be solved using spearman rank correlation

$\boxed{\text{when } X \uparrow Y \uparrow \text{ it should } 1}$

$$\boxed{r_s = \frac{Cov(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))} = 1}$$

| x | y | X <br> R(x) | Y <br> R(y) |
|---|---|---|---|
| 1 | 2 | 2 | 1 |
| 3 | 4 | 3 | 2 |
| 5 | 6 | 4 | 3 |
| 7 | 8 | 5 | 5 |
| 0 | 7 | 1 | 4 |

} rank on least to highest data.